

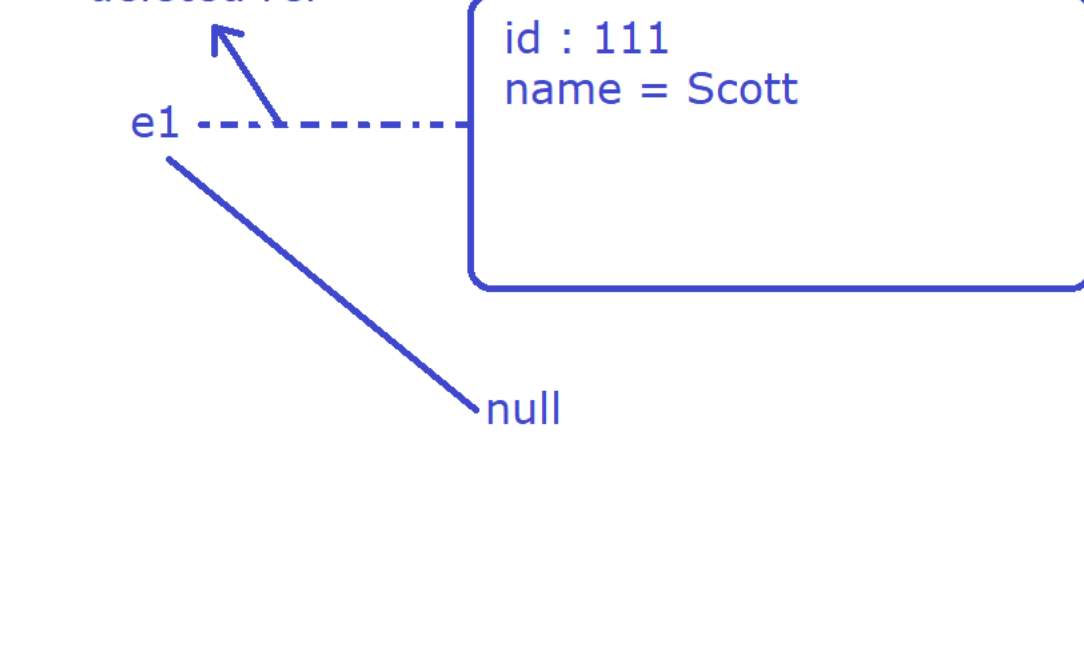
How many ways we can make an object eligible for Garbage Collector :

Basically, there are 3 ways to make an object eligible for Garbage Collector

1) To assign a null literal on reference variable :

```
Employee e1 = new Employee(111,"Scott");  
  
e1 = null;
```

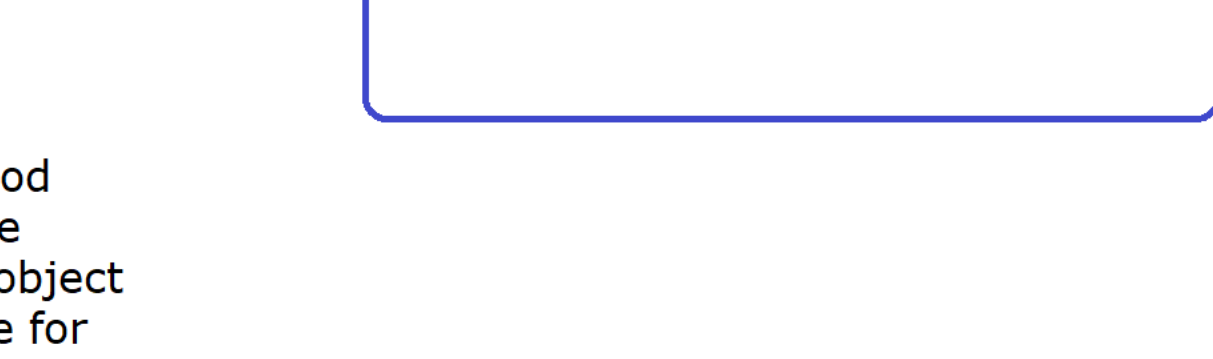
e1 was pointing to 1000x but now e1 is pointing to null, as a result Employee Object created at 1000x memory location is eligible for Garbage Collector



2) Creating an Object inside a method :

```
public void createObject()  
{  
    Student s1 = new Student(1, "Raj");  
}
```

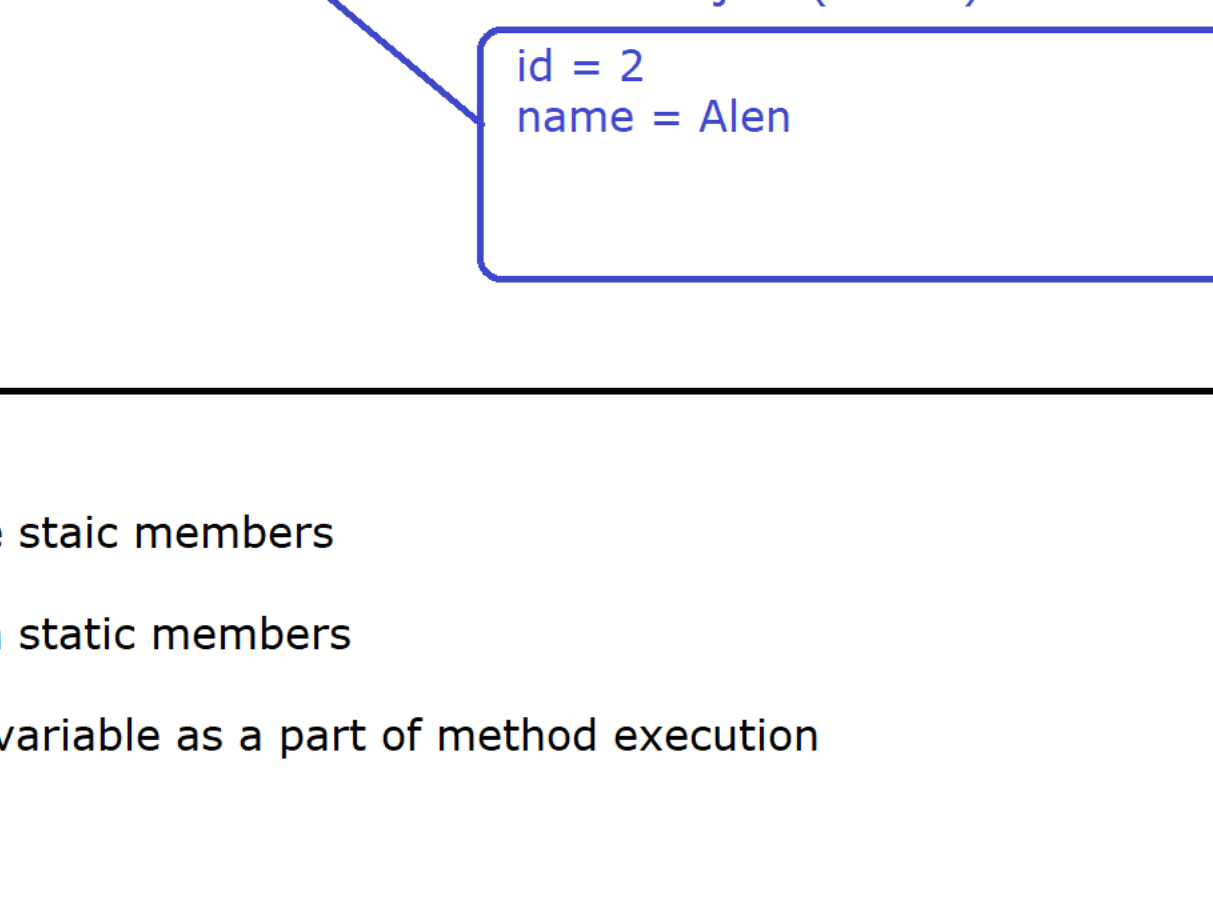
Note : s1 is a local reference variable so once the method execution is over then s1 will be deleted from the corresponding Stack Frame so now the Student object created at 2000x memory location will be eligible for GC



3) Assigning another object on the existing reference variable :

```
Person p1 = new Person(1, "John");  
  
p1 = new Person(2, "Alen");
```

Note : Earlier, p1 reference variable was pointing to Person object created at 1000x memory location but now p1 reference variable is pointing to 2000x memory location (new object) so, 1000x object is eligible for GC



Different types of memories in Java :

- 1) Method Area OR Class Area : Responsible to hold the static members
- 2) HEAP Area : Responsible to hold the objects i.e. non static members
- 3) Stack Area : Responsible to hold local & parameter variable as a part of method execution

****HEAP & STACK Diagram :**

* After learning HEAP & STACK Diagram
a) Complete Object Visualization
b) How many Objects are eligible for GC
c) Output of any complex program

Example :

public class Student
{
 int id = 100;

 public static void main(String []args)
 {
 int x = 500;
 Student s1 = new Student();
 s1 = new Student();
 }
}

Student.java

javac Student.java
java Student

HEAP MEMORY

1000x : StudentObject, id : 100
2000x : StudentObject, id : 100

STACK MEMORY

main_stack

x : 500
s1 : ~~1000x~~ 2000x

When we execute a java program, JVM will get some memory from the Operating System, that memory is divided into two parts

- 1) HEAP AREA [Objects]
- 2) STACK AREA [Method]

Heap and Stack Diagram for CustomerDemo.java

HEAP MEMORY

1000x : CustomerObject, name :2000x, , id :~~2~~ 5
2000x : StringObject, Ravi

3000x : CustomerObject, name : 4000x, id : ~~7~~ 9
4000x : StringObject, Rahul

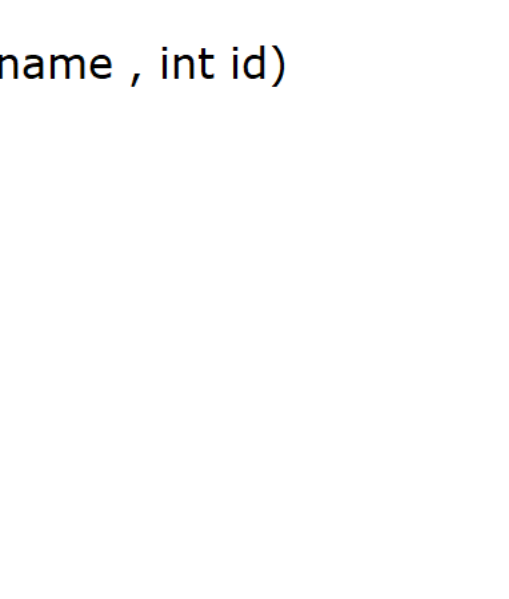
Output :

9
5

STACK MEMORY

main_stack

val : 100
c : 1000x



m1 method execution is over so deleted from Stack memory

```
class Customer  
{  
    private String name;  
    private int id;  
  
    public Customer(String name , int id)  
    {  
        super();  
        this.name=name;  
        this.id=id;  
    }  
  
    public void setId(int id)  
    {  
        this.id=id;  
    }  
  
    public int getId()  
    {  
        return this.id;  
    }  
}  
  
public class CustomerDemo  
{  
    void main()  
    {  
        int val = 100;  
  
        Customer c = new Customer("Ravi",2);  
  
        m1(c);  
  
        //[1 object created at 3000x is eligible for GC]  
  
        IO.println(c.getId());  
    }  
  
    public void m1(Customer cust)  
    {  
        cust.setId(5);  
  
        cust = new Customer("Rahul",7);  
  
        cust.setId(9);  
        IO.println(cust.getId());  
    }  
}
```